Asimov = median of toy pseudo-experiments?

Checks for histogram & NSBI-based test statistics

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Introduction

arXiv:1007.1727

- Analyses commonly rely on the Asimov dataset to extract confidence intervals. When certain conditions hold (e.g. asymptotic regime), the single Asimov fit should equal the median of many toy fits.

Asimov dataset:

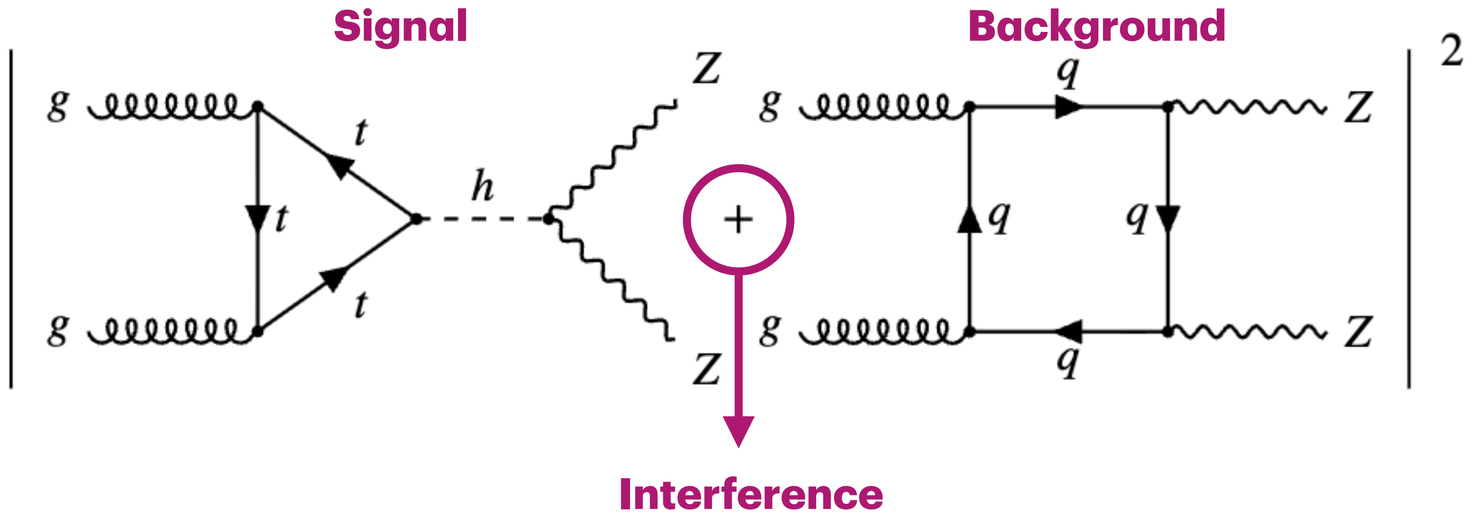
- One dataset, no statistical fluctuation.
- Every event weight set to its exact expectation value.
- Fit it once, one NLL (negative log-likelihood) curve, one $\hat{\mu}$.

Toy pseudo-experiments:

- Many pseudo-experiments.
- Every event weight Poisson-fluctuated around its expectation.
- Fit each one separately from a distribution of NLL curves and $\hat{\mu}$ values.
- Very computationally expensive

> When does Asimov = median of toys actually holds or breaks?

Process of study: not HWW... :(



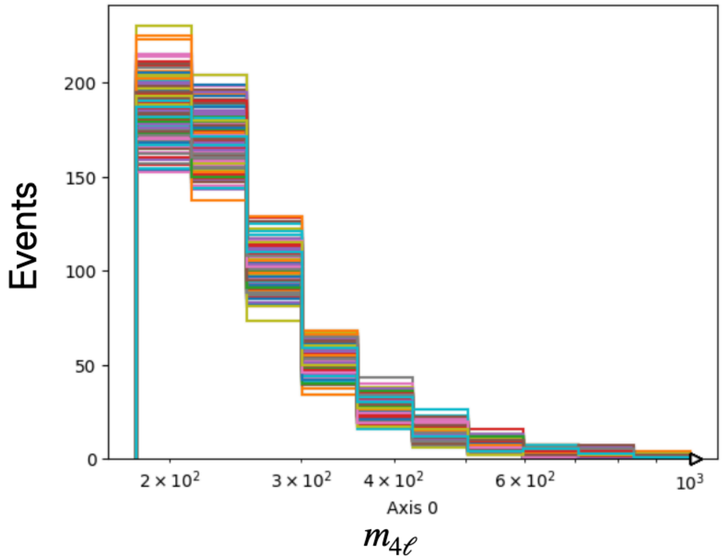
Expected events at $\mu = 1$

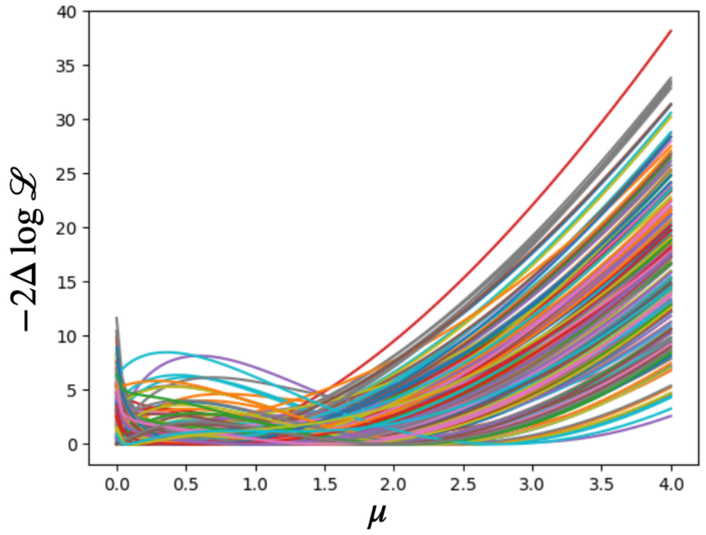
	1 fb^{-1}	100 fb^{-1}	1000 fb^{-1}
signal	0.31	31.0	309.9
background	5.95	595.3	5953.0
interference	-0.61	-60.6	-606.4
total	5.66	565.7	5656.5

- Process: $gg \to H^* \to ZZ \to 4\ell$
- Parameter of interest: Signal strength μ

Setup

- Histogram analysis: $m_{4\ell}$ histogram with 10 logarithmic bins in [180, 1000] GeV
- Toys: 2000 toys per luminosity scanned from 1 to 3000 fb^{-1}
- Fit parameters: only μ scanned over [0, 4]. No nuisance parameters





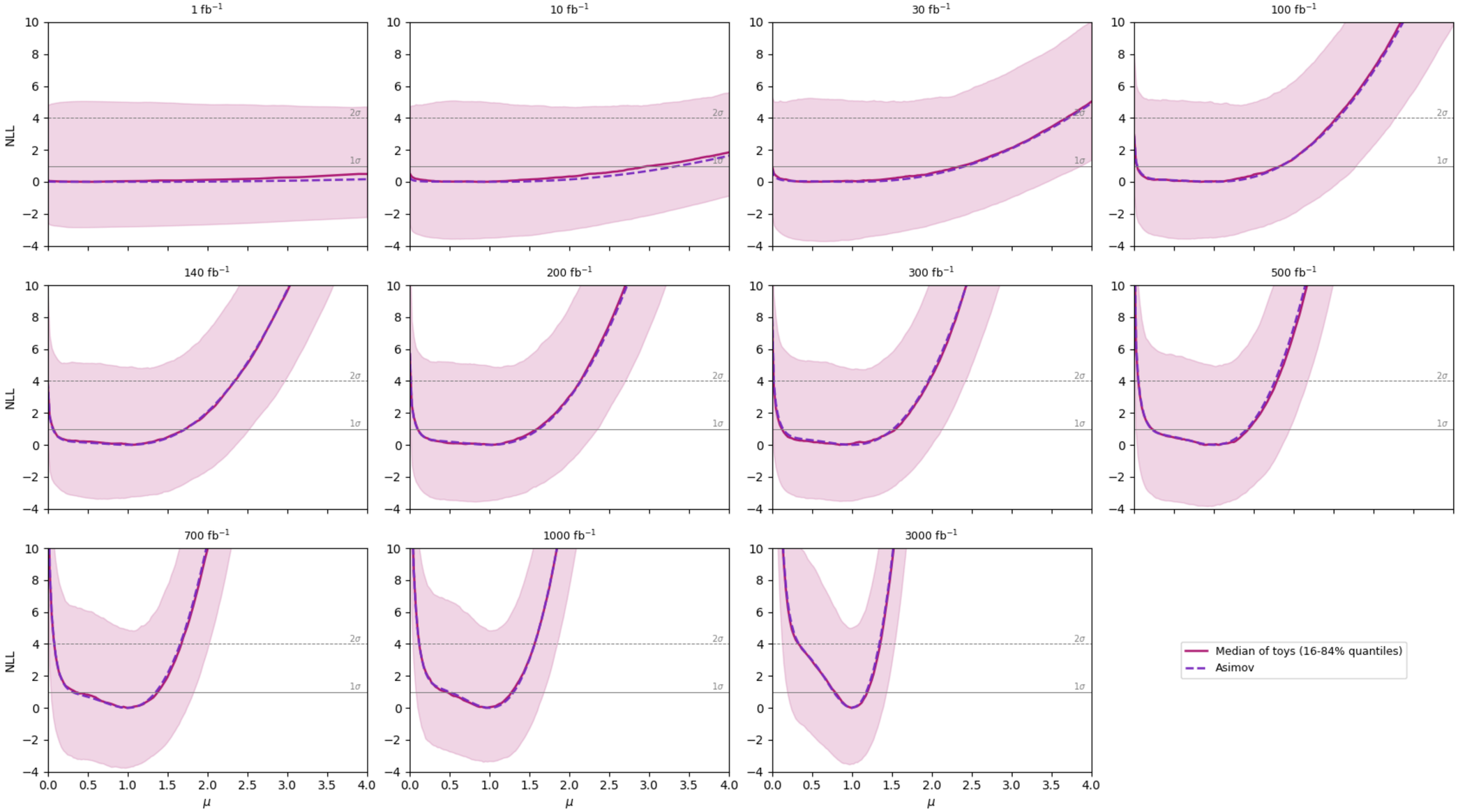
Problem: How to take the median?

Tae's version (his intuition):

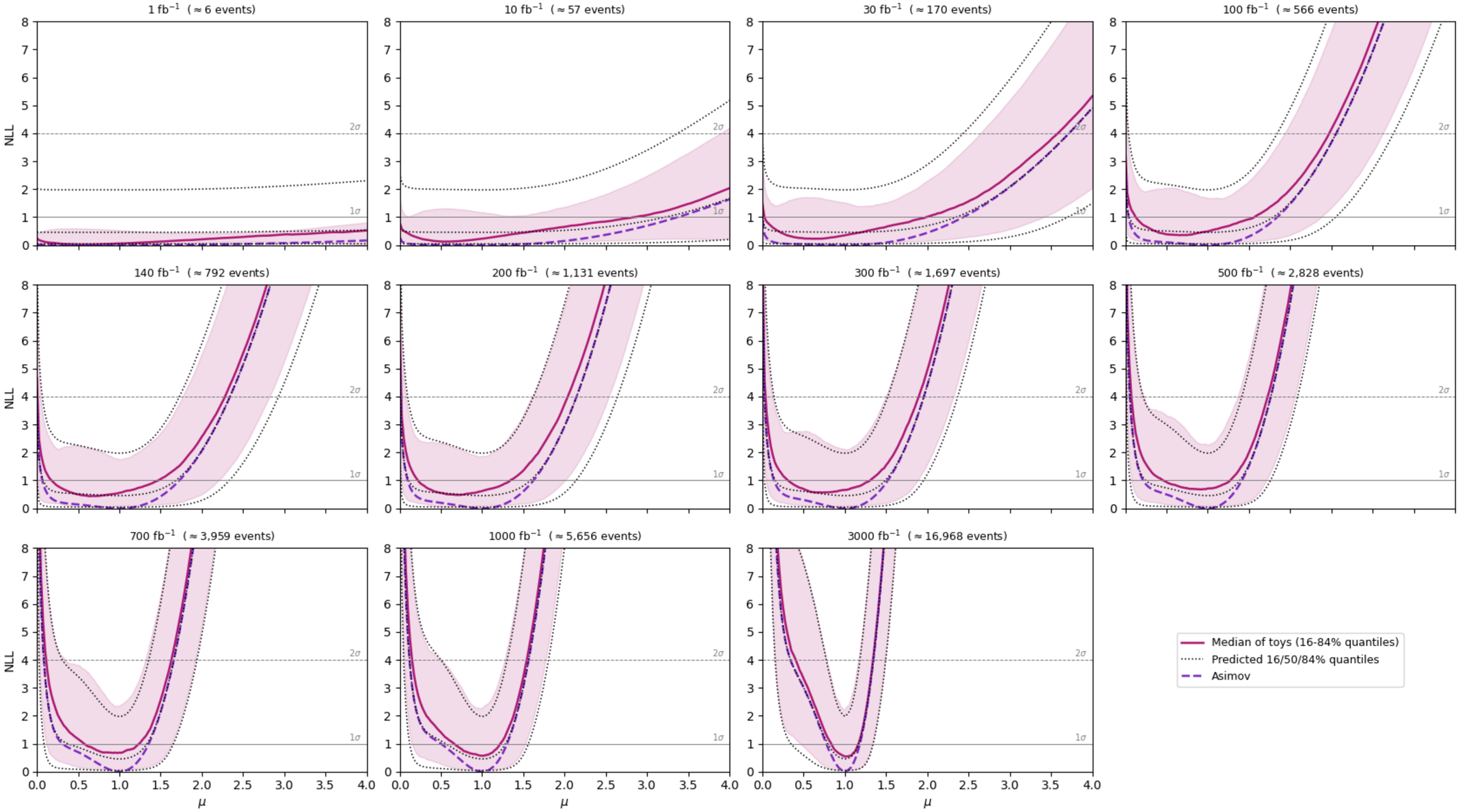
- Not subtract the best-fit value for each toy.
- At least part of the offset comes from the amount of event yield (i.e. lumi value).
- Without subtracting anything off, both between Asimov and toys, the toys should converge to the Asimov *with* this arbitrary offset. Hence, I "should" do the minimum subtraction only after taking the median of the toys.

- I thought we were supposed to first subtract the best-fit value for each toy.
- None of us knows for sure which procedure is correct...
- But they yield very different results and conclusions

Results: Tae's version



Results: “My” version



Questions

- How to extract the median ?
- Are the NLL actually supposed to agree or just the quantiles/ extracted CLs ?
- Are these plots the best way to study when this holds/breaks ?
- Because of the signal and background interference is this process a good candidate for the Asimov to hold?